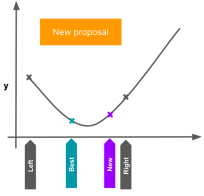
# Optimization in Machine Learning

## Univariate optimization
## Golden ratio



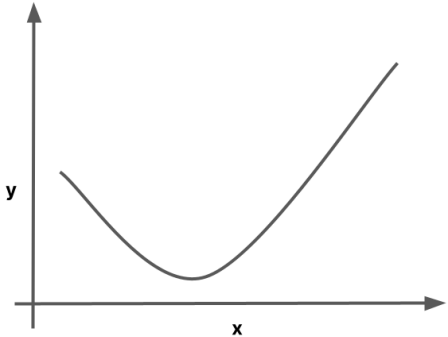

**Learning goals**

- Simple nesting procedure
- Golden ratio

# UNIVARIATE OPTIMIZATION

Let $f : \mathbb{R} \to \mathbb{R}$.

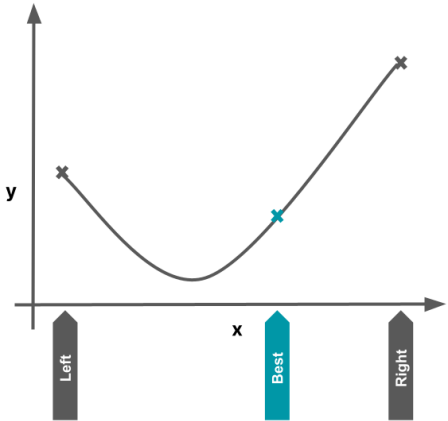
**Goal**: Iteratively improve eval points. Assume function is unimodal. Will not rely on gradients, so this also works for black-box problems.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

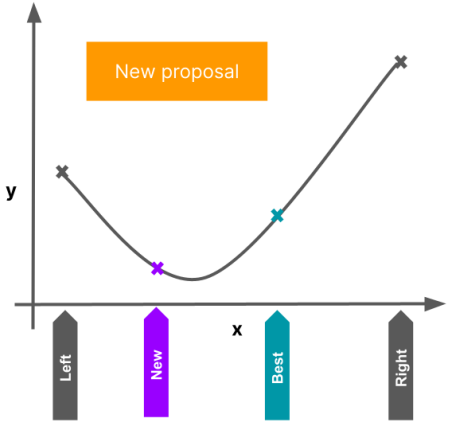
Always maintain three points: left, right, and current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

Propose random point in interval.

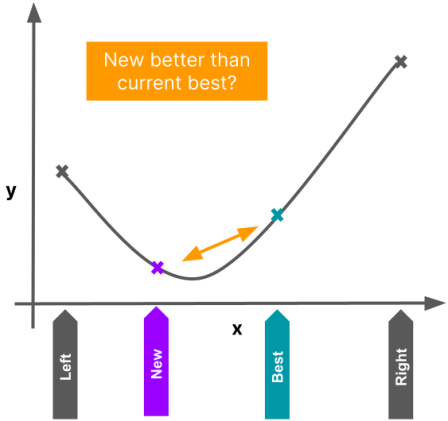



NB: Later we will define the optimal choice for a new proposal.

# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

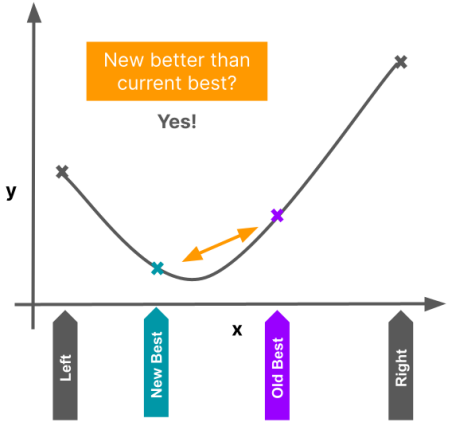
Compare proposal against current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

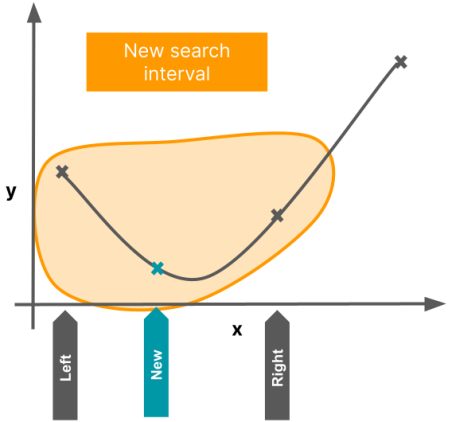
If it is better: proposal becomes current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

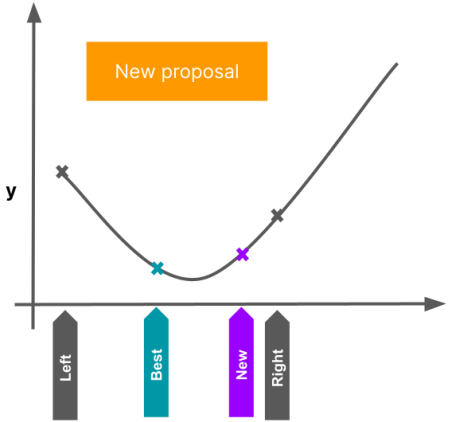
New search interval: around current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

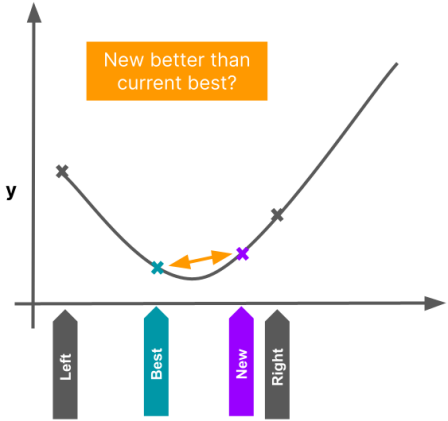
Propose a random point.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

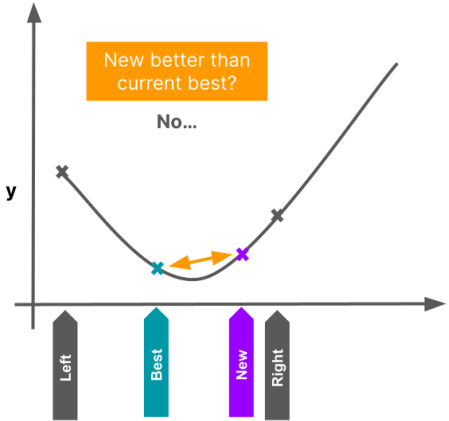
Compare proposal against current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

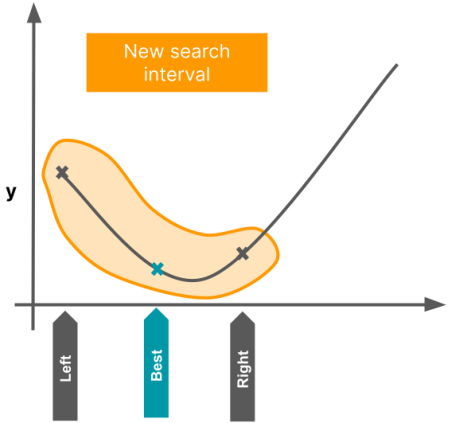
If it is better: proposal becomes current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

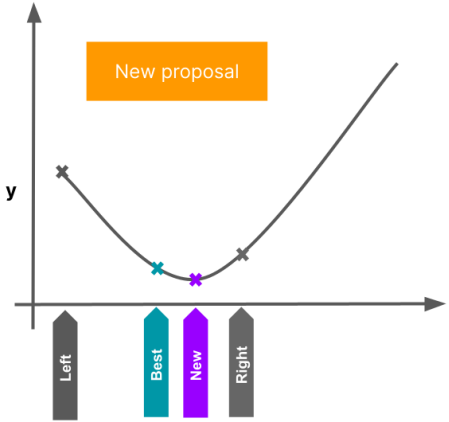
New search interval: around current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

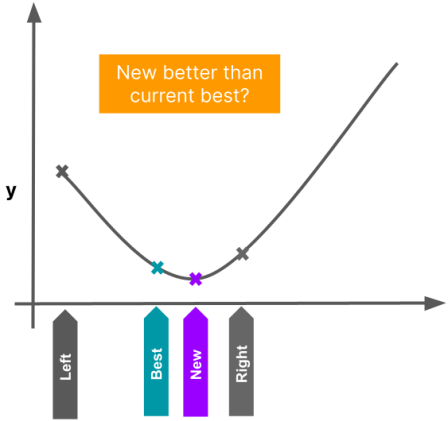
Propose a random point.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

Compare proposal against current best.



# SIMPLE NESTING PROCEDURE

Let $f : \mathbb{R} \to \mathbb{R}$.

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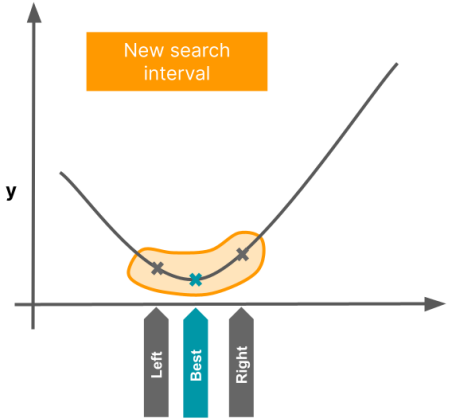
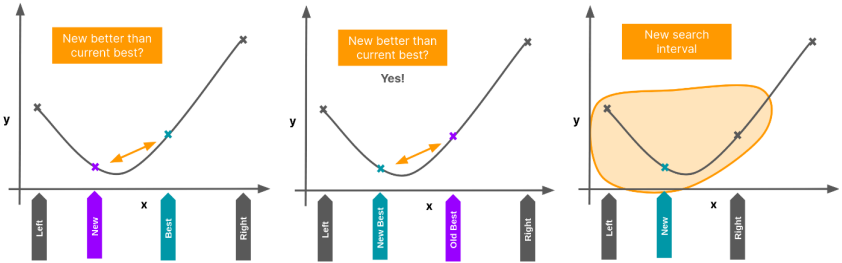
New search interval: around current best.



# SIMPLE NESTING PROCEDURE

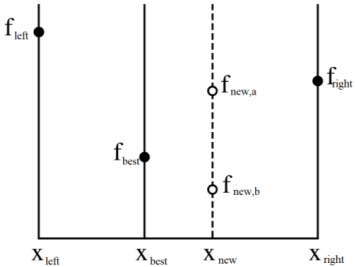
- **Initialization**: Search interval $(x^{\text{left}}, x^{\text{right}})$, $x^{\text{left}} < x^{\text{right}}$
- Choose $x^{\text{best}}$ randomly.
- For $t = 0, 1, 2, \ldots$
  - Choose $x^{\text{new}}$ randomly in $[x^{\text{left}}, x^{\text{right}}]$
  - If $f(x^{\text{new}}) < f(x^{\text{best}})$:
    - $x^{\text{best}} \leftarrow x^{\text{new}}$
  - New interval: Points around $x^{\text{best}}$



# GOLDEN RATIO

**Key question:** How can $x^{\text{new}}$ be chosen better than randomly?

- **Insight 1:** Always in bigger subinterval to maximize reduction.
- **Insight 2:** $x^{new}$ symmetrically to $x^{best}$ for uniform reduction.




Consider two hypothetical outcomes $x^{\text{new}}$: $f_{new,a}$ and $f_{new,b}$.